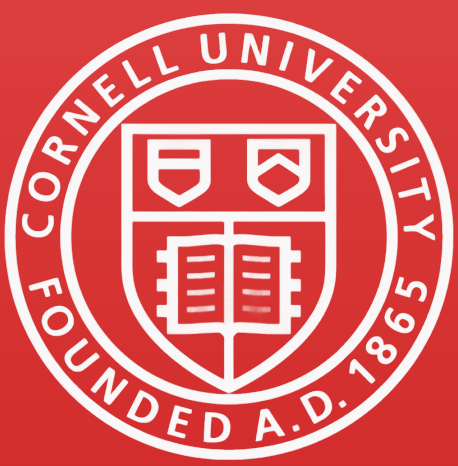




Tree of Thoughts: Deliberate Problem Solving with Large Language Models

CS 4782 Final Project

Aaron Gao (apg86, Cornell University College of Arts & Sciences) | Ashlie Zhang (az468, Cornell University College of Engineering)
Brian Liu (bjl229, Cornell University College of Agriculture & Life Sciences) | Ethan Huang (ejh249, Cornell University College of Arts & Sciences)

Introduction

Motivation

- Background:** LLMs follow a general token-level left-to-right reasoning process
- Problem:** LLMs can only explore one linear line of thinking with no ability to backtrack, even if it is incorrect
- Solution:** Tree of Thoughts (ToT) addresses this issue by exploring multiple reasoning paths simultaneously, enabling backtracking

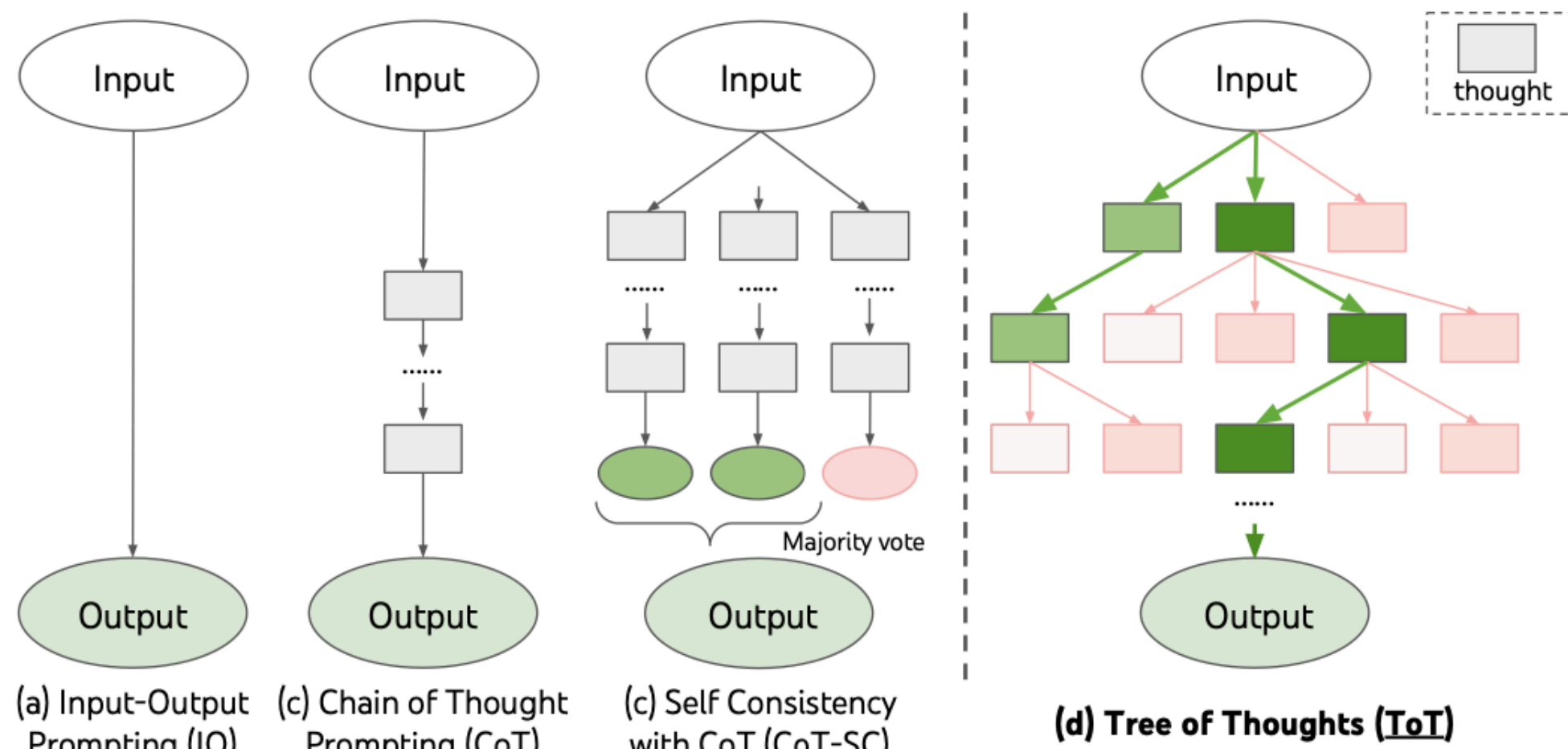


Figure 1 - Schematic illustrating various approaches to problem solving with LLMs. Source: *Tree of Thoughts: Deliberate Problem Solving with Large Language Models*

Our Goal

- Replicate the Game of 24 experiment with other LLM models, validating ToT's significantly higher success rate over CoT-SC, CoT, and IO prompting
- Explore other avenues of tree exploration

Method	Success
IO prompt	7.3%
CoT prompt	4.0%
CoT-SC (k=100)	9.0%
ToT (ours) (b=1)	45%
ToT (ours) (b=5)	74%

IO + Refine (k=10)	27%
IO (best of 100)	33%
CoT (best of 100)	49%

Table 1 - Game of 24 Results with GPT-4. Source: *Tree of Thoughts: Deliberate Problem Solving with Large Language Models*

Paper

- ToT maintains a tree of intermediate "thoughts" using BFS/DFS and LLM self-evaluation to enable backtracking and lookahead
- Evaluated using GPT-4 on complex tasks - Game of 24, Creative Writing, and Mini Crosswords
- ToT outperforms other prompting methods on complex tasks

Methodology

- Evaluate on Game of 24, using same dataset as paper
- Run 100 trials for each prompt method
- Prompting styles: IO, IO (best of 100), CoT, CoT (best of 100), CoT-SC (k=100), and ToT MCTS at branching factors 1 and 5
- For robustness, try all prompt methods with Claude Haiku 4.5, GPT-4o-mini, and Qwen-3.5-2B to investigate effects on models of varying sizes and capabilities

Interesting Experiments

- Explore small model sizes due to compute constraints
- Add Monte Carlo tree traversal, giving it some probability to randomly choose a node, over the simple greedy heuristic

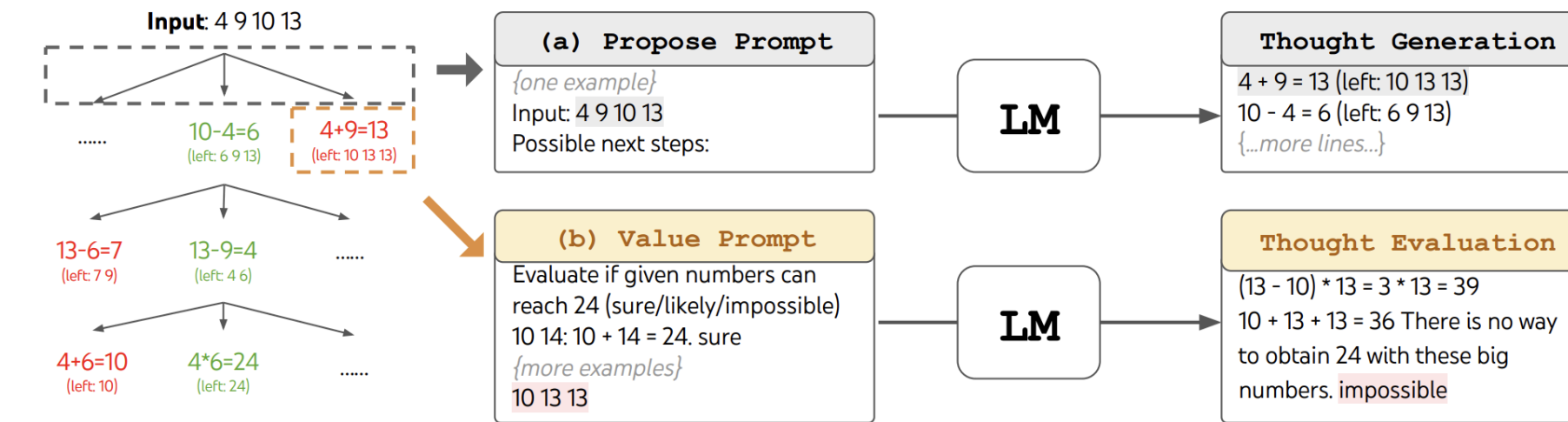


Figure 2 - ToT in a Game of 24. Source: *Tree of Thoughts: Deliberate Problem Solving with Large Language Models*

Conclusion

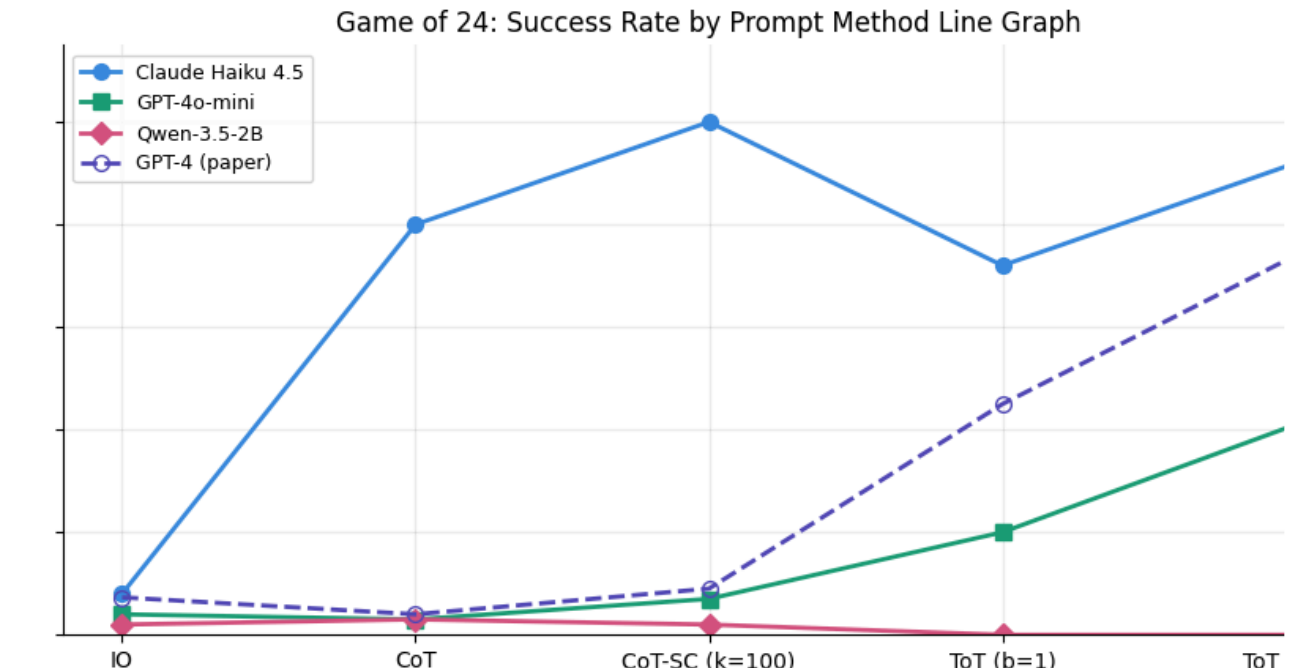


Figure 5 - Line Graph of Success Rates of Different Models by Prompt Method

- Model strength is significantly more important than reasoning strategy
 - ToT fails on very small models (Can't follow the prompt format)
 - ToT fails on strong models (Branches becoming red-herrings)
- ToT with Monte Carlo was able to improve over regular ToT (Heuristic)

Future Work

- Tree traversal with other heuristics & MT w/ self consistency and best of k
- Model in between GPT-o4-mini and Qwen-3.5-2B to find exact point where these methods stop working
- Determine what technique evolution led to ToT performing worse than CoT
- Other games in paper or new games that require different/more reasoning

Results

A. Claude Haiku 4.5

Method	Success
IO	8%
CoT	80%
CoT-SC (k=100)	100%
ToT (b=1)	72%
ToT (b=5)	92%

Table 2 - Game of 24 Results with Claude Haiku 4.5

C. Qwen-3.5-2B

Method	Success
IO	2%
IO (best of 100)	58%
CoT	3%
CoT (best of 100)	69%
CoT-SC (k=100)	2%
ToT (b=1)	0%
ToT (b=5)	0%

Table 4 - Game of 24 Results with Qwen-3.5-2B

- Model does better on CoT-SC, ToT adds too much noise and complexity

- Model fails on ToT → not strong enough

A, B, & C Visualization

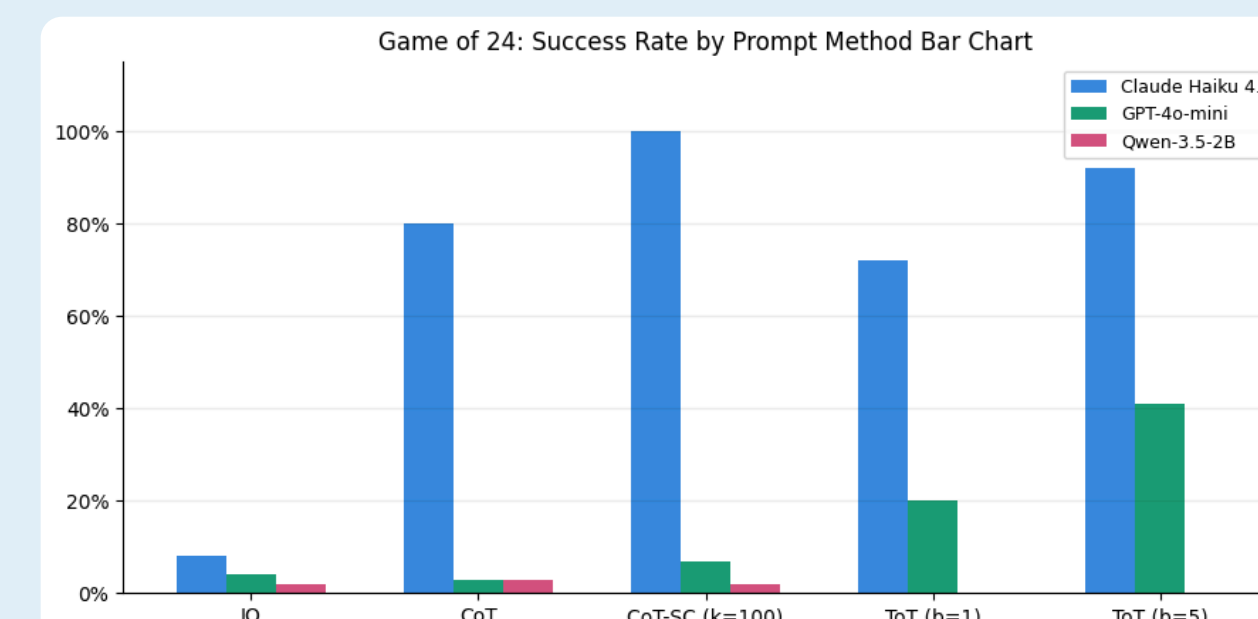
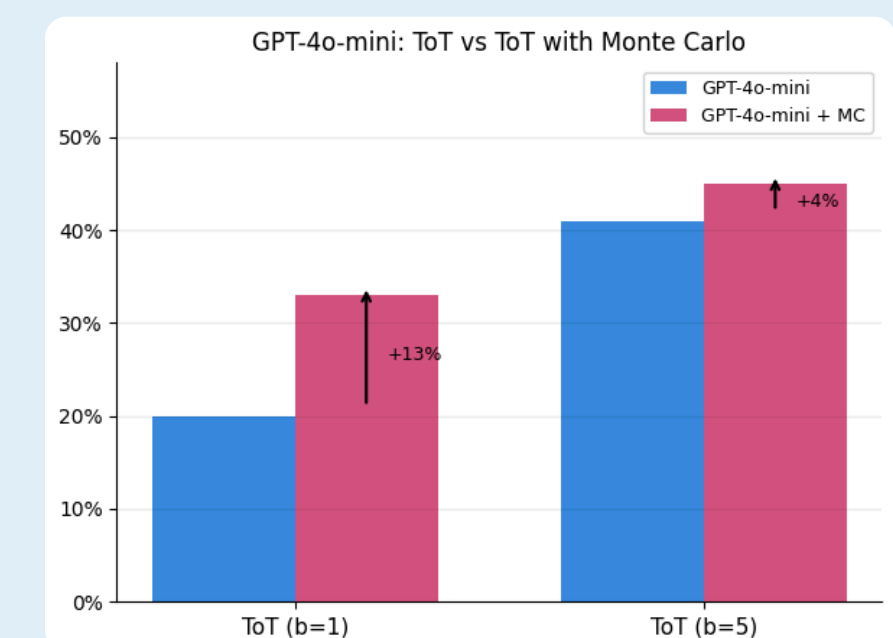


Figure 3 - Bar Graph of Success Rates of Different Models by Prompt Method

D. Monte Carlo ToT with GPT-4o-mini

Method	Success
ToT (b=1)	20%
ToT-MC (b=1)	33%
ToT (b=5)	41%
ToT-MC (b=5)	45%



Left: Table 5 - Game of 24 Results with GPT-4o-mini for ToT with Monte Carlo and Without

Right: Figure 4 - Comparison of Success Rates with Monte Carlo in ToT

- Visible improvement in success rate with ToT with Monte Carlo

References

Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., & Narasimhan, K. (2023). Tree of thoughts: Deliberate problemsolving with large language models. *Advances in Neural Information Processing Systems* (NeurIPS 2023).